

What Musicians Need: Design Principles for VR-Based Performance Preparation to Cope with Performance Anxiety

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Abstract

Music performance anxiety (MPA) affects musicians across experience levels and can significantly undermine performance quality and well-being. While digital technologies show promise to support MPA, it remains unclear how immersive systems can be designed to align with musicians' performance preparation practices. To address this gap, we conducted semi-structured interviews with 20 music practitioners to examine existing coping strategies and expectations for virtual reality (VR)-based support. Our findings identify key challenges and needs articulated by musicians, which inform four design principles addressing contextual diversity, stage-centred immersive performance simulation, integrated skill-anxiety development, and usability for practice compatibility. This work lays the foundation for the design and future deployment of VR-based practice tools and for advancing design knowledge for immersive performance preparation systems.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **User centered design**; • **Applied computing** → **Performing arts**.

Keywords

Music performance anxiety, Virtual reality, Performance preparation, User-centred design, Design principles

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1 Introduction

Music performance anxiety (MPA) is a prevalent challenge experienced by musicians across training levels and professional contexts, with substantial consequences, including reduced performance quality, motivation, career longevity and well-being [1, 10, 14]. Although targeted interventions can support anxiety regulation and performance stability, many established approaches are difficult to integrate into everyday training due to their reliance on specialised expertise, structured programs, or clinical access [4, 13].

Virtual reality (VR) has emerged as a promising medium for supporting performance preparation, as it enables the simulation of performance situations that are difficult to rehearse repeatedly in real-world settings [3, 9, 21]. While prior work demonstrates the feasibility of VR-based exposure, it offers limited guidance on how immersive systems can be grounded in musicians' everyday practice routines and embodied performance experiences.

To address this gap, we adopt a user-centred design approach to examine how music practitioners cope with MPA and how VR could support realistic performance preparation. We conducted semi-structured interviews with 20 music practitioners, informed by prior questionnaires, and supported by structured visual materials. Using thematic analysis, we identified core user needs and translated them into design principles for VR-based performance preparation systems for coping with MPA.

This study contributes an empirically grounded understanding of musicians' needs for VR-supported performance preparation and four design principles addressing contextual diversity, stage-centred immersive simulation, integrated skill-anxiety development, and usability for practice compatibility.

2 Related Work

Digital technologies are increasingly used in various fields where practitioners aim to optimise their performance under evaluative or high-pressure conditions [7, 22]. VR is particularly promising because it enables the simulation of performance situations that are difficult or impractical to rehearse repeatedly in the real world [19]. In the context of MPA, this capacity for controlled simulation makes VR a compelling medium for performance preparation.

Prior research on VR interventions for MPA has primarily focused on simulating evaluative performance contexts. Existing systems commonly draw on virtual reality exposure therapy (VRET) approaches to recreate typical evaluative environments such as concert halls, audition rooms, or examination spaces, and incorporate performance elements including virtual audiences, judges, or stage layouts to elicit feelings of observation and evaluation [2, 3, 12, 16, 23]. Some studies extend environmental simulation by integrating additional support mechanisms, such as guided relaxation or breathing exercises [21] or performance-related feedback [9]. These approaches are typically grounded in exposure-oriented training, enabling musicians to repeatedly engage with performance pressure in controlled environments.

Although prior studies demonstrate the feasibility of VR-based exposure for MPA, they primarily focus on recreating and varying evaluative performance situations to elicit and regulate anxiety and remain largely technology-driven. While musicians commonly participate in evaluation phases, they are less frequently involved in early-stage requirement generation and system design decisions, such as defining relevant stressors, structuring exposure progression, or shaping feedback strategies [6, 12]. As a result, existing work provides insufficient guidance on how VR systems can be designed to align with musicians' practice routines, embodied instrument interaction, and the interplay between technical demands and anxiety.

For musicians, performance anxiety often involves the interaction between contextual, technical, and performance-related factors [17, 18]. Coping strategies for MPA that focus solely on external exposure conditions may therefore overlook these interacting dimensions. Without grounding system design in musicians' lived experiences, VR tools risk reducing performance preparation to scene replication rather than supporting meaningful rehearsal processes, limiting their practical relevance and long-term adoption. Adopting a human-computer interaction (HCI) perspective, this study addresses this gap by grounding design principles in musicians' lived performance practices. Through a user-centred investigation of coping strategies and expectations for VR support, we derive structured design principles intended to inform the development of immersive systems for music performance preparation.

3 Methodology

3.1 Participants

To address this gap, we recruited 20 music practitioners across diverse roles, including educators, students, amateur musicians, and a hobbyist. Participants varied in age, gender, musical background, instruments, and genres (see Table 1). Recruitment was conducted via social media and supplemented by snowball sampling. All participants were over 18 and had prior public performance experience and regularly engaged in performance preparation. Ethics approval was obtained for the project from the university ethics committee. Participation was voluntary and anonymised, and the sample size aligns with established norms for human factors interview studies [5].

3.2 Procedure

We employed a sequential questionnaire-informed interview study, combining an online questionnaire with semi-structured interviews, to explore musicians' coping strategies and expectations for VR-based MPA support. Questionnaire responses informed subsequent interviews, conducted online and face-to-face, which examined how musicians manage MPA in practice and the limitations of existing strategies. To support reflection on potential VR requirements, interviews incorporated structured visual materials tailored to participants' prior familiarity with immersive technologies. These materials comprised a brief slide-based introduction to immersive technologies and their role in music performance preparation, along with screenshots and short video clips of existing applications presented as illustrative examples. All interviews were audio-recorded and transcribed verbatim. The interview procedure and prompts were developed with the co-authors. Interviews were conducted by the first author, with feedback from the co-authors provided throughout the data collection and analysis process.

3.3 Data Analysis

Data were analysed through the thematic analysis [11] using NVivo 15. Inductive coding identified patterns in participants' strategies, challenges, and expectations for technology-enhanced support. The coding process was iterative and interpretive, leading to a structured codebook that informed the development of design principles for a user-centred VR framework. Additionally, three co-authors independently reviewed the themes and subthemes to confirm that they were coherent, distinct, and conceptually consistent.

4 Findings

4.1 User-centred Requirements for the VR Strategy for MPA

4.1.1 UCR1: Adaptive Practice Capability Through Contextual Diversity. Adaptive practice requires opportunities to rehearse across diverse performance setups to build confidence, predictability, and situational readiness. Participants described regular rehearsal and simulated performances as key strategies for preparing for evaluative situations, such as mentally rehearsing stage entry and inviting others to act as an audience. Through repetitive processes, musicians internalise both technical demands and contextual conditions, gradually transforming nervousness into routine. However, sustaining such practices is often constrained by limited access to appropriate venues, equipment, personnel, and time. Participants noted that travelling to performance spaces and securing rehearsal slots can be time-consuming and impractical within everyday routines. As a result, opportunities to repeatedly practice within specific performance environments are unevenly distributed and difficult to maintain.

Musicians emphasised the importance of becoming familiar with particular performance settings in advance, as this familiarity enables them to quickly regain a prepared state when encountering similar environments in real performances. As one participant noted, "...get familiar with specific performance environments, and then continue to practice in those environments, and then when

Table 1: Participant Demographics and Background Information

No.	Age	Gender	Highest Education	Musician Type	Training Years	Started Age	Practice Frequency	Instruments	Immersive Tech Familiarity
P1	25–34	Female	Master	Music educator	>10	Under 6	Few/week	Keyboard	3
P2	18–24	Female	High School eq.	Music student	1–3	13–18	Few/week	Keyboard	5
P3	25–34	Female	Master	Music educator	7–10	13–18	Weekly	Vocal	3
P4	25–34	Female	Bachelor	Music educator	>10	Under 6	Few/week	Keyboard, String, Vocal	1
P5	25–34	Female	Master	Music educator	>10	13–18	Few/month	Vocal	3
P6	25–34	Female	Master	Music educator	>10	Under 6	Weekly	Vocal	5
P7	25–34	Female	Doctoral	Amateur musician	7–10	7–12	Rarely	String	9
P8	25–34	Female	Master	Music educator	>10	Under 6	Few/month	Keyboard, String	0
P9	25–34	Male	Master	Music educator	>10	Under 6	Few/month	Woodwind	0
P10	18–24	Male	High School eq.	Music student	>10	7–12	Daily	Keyboard	3
P11	25–34	Female	Bachelor	Music student	7–10	13–18	Few/week	Keyboard, String, Percussion, Vocal	0
P12	25–34	Female	Bachelor	Music educator	7–10	7–12	Few/month	Keyboard	1
P13	25–34	Female	Master	Music student	1–3	7–12	Few/month	Keyboard	3
P14	25–34	Female	Master	Amateur musician	4–6	Under 6	Few/week	String, Percussion	1
P15	25–34	Female	Master	Music educator	>10	Under 6	Daily	Keyboard, Vocal, Conducting	1
P16	25–34	Female	Bachelor	Music educator	>10	Under 6	Few/month	Keyboard	0
P17	25–34	Male	Master	Amateur musician	>10	19+	Few/week	Keyboard, Vocal	5
P18	25–34	Female	Bachelor	Music educator	>10	Under 6	Daily	Keyboard, Vocal, Conducting	3
P19	25–34	Female	Master	Hobbyist	4–6	Under 6	Rarely	Woodwind	1
P20	18–24	Female	Bachelor	Music student	>10	Under 6	Daily	Keyboard	2

Note. Immersive technology familiarity was self-reported on a 0–10 scale (0 = no prior experience, 10 = extensive experience).

you really enter that environment in real life, you can immediately find the state you practiced."

Real-world performances span a wide range of configurations, including small recital rooms, large concert halls, live music venues, and classroom examination settings. VR has the potential to support adaptive practice by enabling musicians to rehearse across diverse and realistic performance contexts without the logistical constraints of physical access.

4.1.2 UCR2: Multimodal Performance Immersion. Supporting musical performance in VR requires establishing a stage-centred experience that evokes both situational presence and performance-related emotional engagement. Participants emphasised that VR should recreate realistic performance environments, visually, acoustically, and emotionally, to help adapt to real stages and manage stressors. Achieving this requires the integration of visual, auditory, tactile, and emotional cues that together support an embodied experience of performing.

Music instrument interaction was identified as central to this multisensory coherence, requiring alignment between visual representation, tactile response, and sound output. Participants described how playing a keyboard without a visible instrument in the visual scene felt strange, as the absence of the instrument conflicted with their embodied expectations of performance. Musicians also reported the importance of adapting to instrument touch and tone and venue-specific acoustics, which shape both technical execution and perceptual familiarity.

Beyond instrument interaction, stage-based immersion depends on environmental and social cues. Participants described the need to adapt to stage lighting conditions, audience layout and spatial positioning from the stage perspective. Emotional engagement was closely tied to the perceived presence of audiences, which contributed to emotional engagement by making the experience feel closer to a real performance context, "...it would be more like the feeling of being on stage, because there would be more eyes staring at me..."

4.1.3 UCR3: Integrated Skill-Anxiety Co-Development. Musicians require support for the integrated development of technical competence and anxiety regulation during performance, as these two aspects are tightly interdependent. Interview data showed that technical difficulty, unfamiliar repertoire, and concerns about accuracy, tone, or bodily control often trigger anxiety. At the same time, heightened anxiety was described as impairing technical execution even among skilled performers, creating a reinforcing cycle in which technical instability and anxiety mutually exacerbate one another.

Participants described existing practice strategies as insufficient for addressing this interdependence. While musicians commonly rely on self-recording for post-hoc review and adjustment, this approach provides limited support during anxiety moments when technique begins to break down. As one participant explained, "They record the rehearsal process, and after performing once, they review the recording, analyse their own problems, and then try to improve them." As a result, technical issues are recognised only after performance, missing opportunities for in-the-moment adjustment.

Participants expressed a need for being able to observe their own bodily and musical state without interrupting the flow of playing, for example, through awareness of posture, movement, or expressive control. One participant described how being able to see themselves during performance supported this awareness: "When I can see myself, I can recognise the mismatch during the performance. Without interrupting the flow, I become aware that my movement no longer aligns with my intended standard." These findings indicate a user requirement for practice environments that support the co-development of skill stability and anxiety regulation under performance-like conditions.

4.1.4 UCR4: Minimise Technical Friction and Support Performance Demands. VR-based performance practice systems should remain accessible, easy to operate, and compatible with the perceptual and motor demands of musical performance. Participants raised concerns about physical feasibility, including discomfort, motion sickness, restricted movement, and potential disruption to natural playing posture, noting that these issues can interfere with focus and practice flow. Maintaining clear visual access to the instrument,

score, and hand position was identified as essential for practising in VR. As one participant asked, "If I wear those glasses, how should I play the piano? Will my vision be hindered when I play the piano?"

Usability was also highlighted as a critical requirement for sustained use. Musicians emphasised the importance of minimal setup, rapid session initiation, and intuitive interaction workflows that do not add cognitive burden to already demanding practice routines. Participants noted that complex configuration, extensive guidance requirements, and high equipment costs can discourage adoption, particularly when the perceived benefits do not clearly outweigh the effort required.

4.2 Key Design Principles

Prior VR-based approaches to music performance anxiety have largely focused on recreating performance scenes through repeated exposure. While effective for familiarisation, they provide limited support for understanding how anxiety emerges through contextual variation, embodied interaction, and technical demands during practice. Building on musicians' requirements, we reframe immersive design from scene replication toward performance preparation, translating user-centred requirements into design principles that support interpretability, adaptability, and integration with everyday practices.

4.2.1 DP1: Principle of Adaptive Performance Context. This design principle extends contextual diversity beyond scene replication by structuring VR practice to help musicians identify, interpret, and respond to their own anxiety triggers. While prior research has primarily framed VR as an exposure-oriented intervention [8, 15], our findings suggest the need to move beyond scene replication toward adaptive performance contexts.

Building on these findings, VR systems with adaptive performance contexts can decompose contextual variation into interpretable elements such as audience presence, evaluative cues, spatial scale, and timing structure, making their relationship to anxiety and performance explicit. Without adaptive control, VR-based practice risks reducing performance preparation to repetitive scene exposure, offering limited support for understanding why anxiety emerges in specific musical passages, moments, or evaluative conditions.

To support learning from contextual variation, adaptive performance contexts can support structured opportunities for reflection and experimentation. Brief reflective prompts during or immediately after practice can support awareness of coping strategies and their effects, while repeated practice under controlled contextual variation allows musicians to deliberately compare different approaches to anxiety regulation. In this way, VR functions not simply as a space for exposure, but as a means for musicians to build personalised knowledge about what strategies work for them under different performance conditions.

4.2.2 DP2: Principle of Stage-Centred Immersive Performance Simulation. The integration of visual, auditory, and tactile cues supports embodied interaction and skill development [24]. Designing VR for musical performance preparation requires supporting embodied instrument interaction, adaptation to performance conditions, and audience-related emotional engagement. Performance contexts

demand continuous sensorimotor coupling with the instrument while situating performers under evaluative conditions that shape attention and affect.

Instrument-oriented interaction is central to immersive performance simulation. Coherent multisensory experience relies on preserving alignment between performers' actions, visual feedback, tactile contact, and sound output. Where supported by stable passthrough rendering, interaction can be anchored to performers' own physical instruments, enabling direct visual access to real hands and contact surfaces through passthrough rendering. This approach preserves the performer's real hand-instrument interaction while overlaying virtual performance elements such as audience presence and lighting. For example, a pianist may retain visibility of their keyboard and finger movements, while a string or wind player maintains access to bow-string contact and finger placement on keys or valves, without disrupting established motor patterns or bodily coordination. A lack of stage-centred, instrument-oriented immersion can cause VR practice to diverge from real performance conditions, reducing musicians' capacity to adapt embodied interaction, spatial orientation, and evaluative pressure. Maintaining spatial alignment between real and simulated contexts supports smoother transfer from VR rehearsal to live performance.

Stage-based environmental cues further ground performers within a performance context. Lighting and acoustics should be implemented as adjustable rehearsal variables rather than static scene features, allowing performers to practice adapting to brightness, glare, and venue-specific acoustic conditions. Perceiving audience layout and spatial positioning from the stage perspective supports situational orientation.

Audience-driven evaluative cues play an important role in shaping performers' emotional responses during performance. Audience presence should be designed as an active element, with controllable parameters such as size, proximity, gaze behaviour, and temporal dynamics. These cues enable repeatable and graded exposure to evaluative pressure while preserving the emotional character of live performance.

4.2.3 DP3: Principle of Skill-Anxiety Regulation. VR practice environments can support the integrated development of technical competence and anxiety regulation during musical performance. This design reflects how technique and affect interact in real performance. VR can place musicians in performance conditions that support skill refinement while increasing awareness of bodily and emotional states under pressure. Embodied representations, such as body tracking or virtual mirroring, can support awareness of posture, movement, and tension. Multiple perspectives, including performer, third-person, or audience views, can help musicians reflect on changes in technique and stage presence. Subtle visual or haptic cues can convey technical and anxiety-related states without disrupting musical flow.

Feedback plays an important role in learning strategies, supporting improved performance outcomes during skill practice [20]. VR systems can reveal recurring anxiety and technical breakdowns and enable short performance replays from multiple perspectives. Viewing recent passages from performer, audience, and reflective

perspectives helps musicians connect emotional state, bodily control, and musical outcome, fostering transferable self-regulation strategies.

4.2.4 DP4: Principle of Usability and Accessibility. Effective VR systems for musical practice prioritise usability and accessibility to remain compatible with the perceptual, motor, and attentional demands of performance. Excessive physical burden, visual occlusion, or interaction complexity can disrupt posture and focus. VR designs that minimise technical friction and preserve access to notation, instrument surfaces, and hand position better support performance readiness.

Interaction modalities should be designed to preserve natural playing posture and maintain clear visual access to instruments and scores. Passthrough or mixed-reality capabilities support continuous visibility of hands, instrument surfaces, and notation without requiring headset removal. Motion tracking supports a full range of instrumental movements without constraining expressive or technical execution. Cognitive load is reduced through intuitive and streamlined workflows that enable musicians to begin practice quickly. Clear visual and auditory cues support essential tasks, such as selecting venues, adjusting difficulty, and reviewing performance data, without diverting attention from musical performance.

5 Conclusion

This paper investigates how virtual reality can support musicians' performance preparation for coping with music performance anxiety, grounded in musicians' everyday practice experiences. Through interviews with 20 music practitioners, we identified key challenges related to practising across diverse performance contexts, achieving realistic stage-centred immersion, managing the interplay between technical demands and anxiety, and maintaining usability compatible with instrumental performance. These findings highlight practical gaps between existing VR approaches and musicians' real rehearsal needs.

Based on these insights, we contribute four design principles that inform the design of VR systems for performance preparation. The principles outline how VR can support adaptive practice, preserve embodied instrument interaction, integrate skill development with anxiety regulation, and minimise technical friction during rehearsal. This work provides practice-grounded design guidance for HCI researchers and designers seeking to develop immersive systems that better align with musicians' performance preparation practices.

References

- [1] S. Barros, H. Marinho, N. Borges, and A. Pereira. 2022. Characteristics of music performance anxiety among undergraduate music students: A systematic review. *Psychology of Music* 50, 6 (2022), 2021–2043. doi:10.1177/03057356211037263
- [2] D. Bellinger, K. Wehrmann, A. Rohde, M. Schuppert, S. Störk, M. Flohr-Jost, and others. 2023. The application of virtual reality exposure versus relaxation training in music performance anxiety: A randomized controlled study. *BMC Psychiatry* 23, 1 (2023), 555. doi:10.1186/s12888-023-04974-6
- [3] J. Bissonnette, F. Dubé, M. D. Provencher, and M. T. Moreno Sala. 2016. Evolution of music performance anxiety and quality of performance during virtual reality exposure training. *Virtual Reality* 20 (2016), 71–81. doi:10.1007/s10055-016-0287-9
- [4] E. Blair and H. van der Sluis. 2022. Music performance anxiety and higher education teaching: A systematic literature review. *Journal of University Teaching & Learning Practice* 19, 3 (2022), 05.
- [5] K. Caine. 2016. Local standards for sample size at CHI. In *Proceedings of the 2016 ACM Conference on Human Factors in Computing Systems*. 981–992. doi:10.1145/2858036.2858498
- [6] S. Choi, J. Kim, and S. Kim. 2024. Effects of exposure in computer-assisted virtual environments on vocalists' performance anxiety: A quasi-experimental study. *Journal of Rational-Emotive & Cognitive-Behavior Therapy* 42, 3 (2024), 523–543. doi:10.1007/s10942-023-00515-9
- [7] F. Dermody and A. Sutherland. 2015. A multimodal system for public speaking with real-time feedback. In *Proceedings of the 2015 ACM International Conference on Multimodal Interaction*. 369–370. doi:10.1145/2818346.2830601
- [8] Miranda R Donnelly, Renee Reinberg, Kaori I Ito, David Saldana, Meghan Neureither, Allie Schmiesing, Esther Jahng, and Sook-Lei Liew. 2021. Virtual reality for the treatment of anxiety disorders: a scoping review. *The American Journal of Occupational Therapy* 75, 6 (2021), 7506205040.
- [9] Y. Fanger, K. Pfeuffer, U. Helmbrecht, and F. Alt. 2020. PIANX: A platform for piano players to alleviate music performance anxiety using mixed reality. In *Proceedings of the 19th International Conference on Mobile and Ubiquitous Multimedia*. 267–276. doi:10.1145/3428361.3428472
- [10] A. L. Faur, S. Pintea, S. Vaida, and A. N. Opre. 2023. The efficacy of cognitive and behavioral interventions upon music performance anxiety: A meta-analysis. *Psychology of Music* 51, 2 (2023), 357–372. doi:10.1177/03057356221103238
- [11] Robert P Gauthier and James R Wallace. 2022. The computational thematic analysis toolkit. *Proceedings of the ACM on Human-Computer Interaction* 6, GROUP (2022), 1–15.
- [12] J. L. Gómez-Sirvent, A. Fernández-Sotos, A. Fernández-Caballero, and D. Fernández-Sotos. 2024. Assessment of music performance anxiety in a virtual auditorium through the study of ambient lighting and audience distance. *Virtual Reality* 28, 2 (2024), 71. doi:10.1007/s10055-024-00845-7
- [13] D. T. Kenny. 2005. A systematic review of treatments for music performance anxiety. *Anxiety, Stress, & Coping* 18, 3 (2005), 183–208. doi:10.1080/10615800500167258
- [14] M. Lupiáñez, F. D. P. Ortiz, J. Vila, and M. A. Muñoz. 2022. Predictors of music performance anxiety in conservatory students. *Psychology of Music* 50, 4 (2022), 1005–1022. doi:10.1177/03057356211039663
- [15] Jessica L Maples-Keller, Brian E Bunnell, Sae-Jin Kim, and Barbara O Rothbaum. 2017. The use of virtual reality technology in the treatment of anxiety and other psychiatric disorders. *Harvard Review of Psychiatry* 25, 3 (2017), 103–113.
- [16] E. K. Orman. 2003. Effect of virtual reality graded exposure on heart rate and self-reported anxiety levels of performing saxophonists. *Journal of Research in Music Education* 51, 4 (2003), 302–315. doi:10.2307/3345651
- [17] Ioulia Papageorgi. 2022. Prevalence and predictors of music performance anxiety in adolescent learners: contributions of individual, task-related and environmental factors. *Musicae Scientiae* 26, 1 (2022), 101–122.
- [18] Jose L Gomez Sirvent, Desiree Fernandez-Sotos, Roberto Sanchez-Reolid, Francisco de la Rosa López, Alicia Fernandez-Sotos, and Antonio Fernandez-Caballero. 2023. Pre-occupancy evaluation of a virtual music school classroom: Influence of color and type of lighting on music performers. *Building and Environment* 246 (2023), 119899.
- [19] A. Suh and J. Prophet. 2018. The state of immersive technology research: A literature analysis. *Computers in Human Behavior* 86 (2018), 77–90. doi:10.1016/j.chb.2018.04.019
- [20] K. Sunday, S. S. Anvari, Y. Li, J. Sun, R. R. Wehbe, H. Neyedli, and M. D. Barrera Machuca. 2025. The importance of performance feedback in virtual reality (VR) training: A table tennis coach perspective. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7. doi:10.1145/3706599.3720204
- [21] M. van Zyl. 2020. The effects of virtual reality on music performance anxiety among university-level music majors. *Visions of Research in Music Education* 35, 1 (2020), 15.
- [22] J. Wang, H. Yang, R. Shao, S. Abdullah, and S. S. Sundar. 2020. Alexa as coach: Leveraging smart speakers to build social agents that reduce public speaking anxiety. In *Proceedings of the 2020 ACM Conference on Human Factors in Computing Systems*. 1–13. doi:10.1145/3313831.3376564
- [23] A. Williamon, L. Aufegger, and H. Eiholzer. 2014. Simulating and stimulating performance: Introducing distributed simulation to enhance musical learning and performance. *Frontiers in Psychology* 5 (2014), 25. doi:10.3389/fpsyg.2014.00025
- [24] X. Xu, J. Kang, and L. Yan. 2022. Understanding embodied immersion in technology-enabled embodied learning environments. *Journal of Computer Assisted Learning* 38, 1 (2022), 103–119. doi:10.1111/jcal.12597